FSAM — Formula Student

Bill of Materials Management · Procurement · Supply Chain OR

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BOM Manager & Procurement Coordinator · 2025–2026

Team:	FSAM — Formula Student Arts et Métiers Lille
Role:	BOM Manager, Procurement Coordinator
BOM scope:	~450 parts across 18+ assemblies
Budget:	~€20,000 total vehicle cost
Team size:	12–15 students
Competition:	Formula Student Spain (target 2026)

Supply Chain · BOM Explosion · Critical Path · Procurement Optimization

Contents

1 Executive Summary **5**

2 Formula Student — Competition Context **5**
2.1 About the Competition 5
2.2 FSAM — The ENSAM Lille Team 6

3 Bill of Materials — Structure and Management **6**
3.1 BOM Architecture 6
3.2 Scope 7

4 Procurement Coordination **8**
4.1 Lead Time Management 8
4.2 Cost Report Compliance 8
4.3 Version Control and Design Changes 9

5 Academic Connections **9**

6 Current Status and Results **9**

7 Limitations & Roadmap **10**

8 References **10**

1. Executive Summary

Formula Student is one of the most demanding student engineering competitions in the world. Teams design, build, and race a single-seat formula-style car from scratch, judged on dynamic performance, engineering design, cost report, and business plan. **FSAM** (Formula Student Arts et Métiers Lille) fields a team of 12–15 students competing at European-level events.

Hamza Houir Alami serves as **BOM (Bill of Materials) Manager and Procurement Coordinator** — the person responsible for knowing, at any given moment, which of the car's ~450 parts exist, which are on order, what they cost, and whether the team will meet its build timeline.

This is operationally the most critical non-engineering role on the team. A race car that cannot be assembled because a €3 bracket was never ordered is as broken as one with a faulty suspension design. The BOM role sits at the interface between all technical subteams (chassis, powertrain, suspension, braking, electronics) and the team's budget and logistics constraints.

~450	18+	~€20,000	12–15
Parts in BOM	Assemblies tracked	Vehicle budget	Team members

2. Formula Student — Competition Context

2.1. About the Competition

Formula Student (FS) is an international engineering design competition organized independently in each country, with major events at Silverstone (UK), Hockenheim (Germany), and Montmeló (Spain). Student teams submit to five scored categories:

- **Dynamic events:** Acceleration, Skidpad, Autocross, Endurance — pure on-track performance
- **Cost Report:** Every part must be valued at market replacement cost, regardless of how it was sourced. This creates a formalized accounting discipline that mirrors real industrial cost engineering.
- **Design Report:** Technical judges evaluate engineering decisions, tradeoffs, and justifications
- **Business Plan Presentation:** Teams pitch a hypothetical commercialization strategy for the car

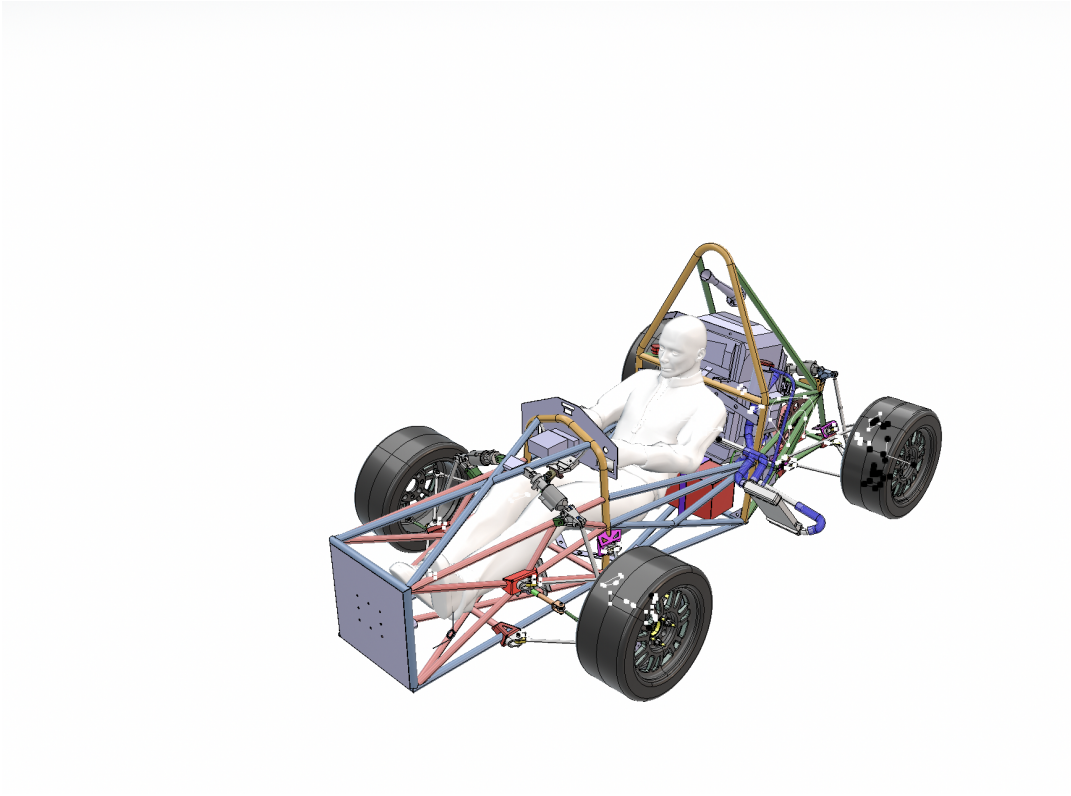

Figure 1: FSAM full vehicle model — 3DExperience (Dassault Systèmes)

2.2. FSAM — The ENSAM Lille Team

FSAM is a fully student-run team operating within ENSAM Lille. The car is designed and built from scratch every year: chassis fabrication, powertrain integration, suspension geometry, braking system, aerodynamic elements, and electronics. The 2025–2026 car uses a **salvaged Honda VFR 800 motorcycle engine**, recovered by the team and integrated into a tubular steel space frame chassis. The engine encountered mechanical issues during integration; the full team worked through these problems together. The build is currently at an advanced stage: chassis complete, engine mounted.

Target competition: **Formula Student Spain** (Montmeló circuit, summer 2026, pending confirmation).

3. Bill of Materials — Structure and Management

3.1. BOM Architecture

A Formula Student car BOM is a **multi-level hierarchical data structure**. The top node is the complete vehicle; below it are major assemblies; below those are sub-assemblies and individual parts. Each leaf entry contains:

Field	Content
Part number	Unique identifier within the BOM tree
Description	Part name and functional role
Assembly	Parent assembly (chassis, suspension, powertrain...)
Subteam responsible	Owner team for that component
Supplier + reference	Vendor and catalog reference
Unit cost / total	Procurement cost + cost report valuation
Lead time	Days from order to delivery
Status	Designed / ordered / received / installed

OR structure: A multi-level BOM is formally a directed acyclic graph (DAG) — each node is a sub-assembly, each edge a dependency, each leaf a purchased part. "Exploding" the BOM to compute gross requirements (total quantities needed at each level) is equivalent to computing a weighted matrix-vector product along all root-to-leaf paths — the same structure as Leontief input-output analysis in macroeconomics.

3.2. Scope

The FSAM BOM covers ~**450 individual part numbers** across **18+ assemblies**, maintained jointly in Excel and 3DExperience. Major assembly groups:

- Chassis (tubular steel space frame, roll hoop, firewall)
- Powertrain (engine, gearbox, chain drive, exhaust)
- Suspension front & rear (uprights, wishbones, push/pull rod, dampers)
- Steering system (rack, column, wheel)
- Braking system (master cylinders, calipers, rotors, pedal box)
- Wheels & tires
- Aerodynamics (nose cone, undertray, optional wings)
- Electronics (ECU, wiring harness, sensors, dashboard)
- Driver safety (harness, seat, helmet support, kill switch)

Figure 2: FSAM chassis — early build stage (space frame, suspension mounts)

4. Procurement Coordination

4.1. Lead Time Management

With ~450 parts and a fixed competition date, procurement timing is a hard constraint. Parts from specialist suppliers (custom bearings, carbon fiber materials, brake hardware) carry 3–6 week lead times. The **critical path** of the build schedule is determined by the longest-lead-time parts on the assembly dependency chain.

Concrete example — engine integration: The team's salvaged Honda VFR 800 motorcycle engine encountered mechanical issues during integration. Resolving these issues was a team-wide effort, requiring diagnosis, component-level intervention, and BOM updates to reflect the modified powertrain specification. This event illustrates the real operational pressure of the BOM role: design changes cascade through part statuses, supplier references, and assembly sequencing simultaneously.

4.2. Cost Report Compliance

Formula Student's Cost Report rules require that **every part be valued at its market replacement cost**, not its actual purchase price. A donated part worth €0 in acquisition cost is still recorded at its €50 market equivalent. This creates a dual-tracking requirement:

Column	Purpose	Notes
Actual spend	Budget tracking	What was paid
Cost report value	FS compliance	Market replacement cost
Delta	Sourcing efficiency	Donated/discounted parts

This rule has an interesting economic rationale: it prevents teams from gaining competitive advantage purely through sponsorship access rather than engineering skill, and ensures the Cost Report reflects real design decisions.

4.3. Version Control and Design Changes

Designs change continuously mid-build. A suspension geometry revision can make 8–12 previously-ordered parts obsolete. BOM version management — tracking which revision of the design each part corresponds to — is one of the most operationally demanding aspects of the role, especially without a PLM system.

5. Academic Connections

Supply chain operations: Managing ~385 referenced parts (transmission, steering, cooling, braking, suspension, electronics...) across 18+ assemblies with real lead times and continuous design changes is the operational analog of a multi-echelon supply system. Key suppliers: SKF and Igus (bearings/guides), Drexler (differential), Taylor Racing (drivetrain), Norelem (standard parts).

Project management under constraints: A fixed competition date, 12–15 students across parallel subteams, and assembly dependencies create a real project scheduling challenge. The Honda VFR 800 engine integration issues were a live example of an unplanned disruption cascading through assembly sequencing.

Cost engineering: The dual cost tracking (actual spend vs. FS market replacement value) builds industrial cost accounting discipline. Aluminium parts (EN AW-2017 A) undergo CNC machining and blue anodisation; steel parts are burnished or painted. Each process choice carries both a performance and a cost report implication.

6. Current Status and Results

450+	18+	Build ongoing
Parts tracked in BOM	Assemblies covered	Competition status

Key milestones reached as of early 2026:

- **Chassis complete** — tubular space frame fabricated and validated
- **Engine sourced and mounted** — motorcycle engine integrated into powertrain assembly after

re-sourcing under Hamza's coordination

- **BOM fully populated** — all 450+ parts entered across 18+ assemblies with cost, supplier, and status tracking
- **Cost report draft prepared** — dual actual/market-value tracking maintained throughout build
- **Key suppliers engaged** — SKF, Drexler, Norelem, Taylor Racing, Igus; lead times tracked across all assemblies

Note: The car is not yet complete as of this writing. Competition participation (FS Spain) is targeted for summer 2026, pending build completion and dynamic testing. Results from dynamic events will be added once available.

7. Limitations & Roadmap

1. **Manual BOM tracking.** The BOM is maintained in Excel without automated synchronization to 3DExperience. A PLM integration would auto-propagate design changes to the BOM — eliminating the largest source of manual error.
2. **No formal schedule model.** Assembly sequencing and deadline tracking are managed informally across the team. A structured project timeline (Gantt, milestones) with explicit dependencies would surface critical path risks earlier.
3. **Manual lead time tracking.** Order timing is managed heuristically (order early if uncertain). A systematic lead time log per supplier would improve precision and reduce both over-ordering and late-delivery risk.
4. **Single-point-of-failure on BOM.** With one BOM manager, any gap in availability creates an information black hole for the team. Distributing BOM maintenance with clear handoff protocols would improve resilience.

Roadmap:

- **M1–M2:** Complete powertrain integration and braking assembly; finalize cost report
- **M3:** Begin dynamic testing; validate BOM against physical build
- **M4–M5:** Competition preparation; resolve any outstanding procurement gaps
- **Post-competition:** Implement a proper digital BOM with automated status tracking for next season

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Hamza Houir Alami, ENSAM Lille, 2026